

W18e - Heat Capacity Ratio $\gamma = \frac{c_p}{c_v}$

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1 Introduction

The objective of experiment W18e is to determine the heat capacity ratio

$$\gamma = \frac{c_p}{c_v}$$

of air and to compare the experimental results with the theoretical expectation. The quantity γ characterizes how a gas responds to compression and expansion and plays a central role in thermodynamics and gas dynamics. In particular, it enters the description of *adiabatic* processes, where heat exchange with the surroundings is negligible. Such processes are relevant whenever changes occur rapidly, for example during the propagation of sound waves in gases.

In this experiment, γ is determined using two independent approaches. In the first part, the speed of sound v in air is measured with an ultrasound transmitter–receiver setup mounted on an optical bench. The wave is generated by a signal generator and detected as a voltage signal at the receiver, which is recorded with an oscilloscope. The sound velocity is obtained (i) from the phase shift between transmitter and receiver signal using Lissajous figures in the oscilloscope's x – y mode, and (ii) from a transit-time measurement using pulsed excitation (burst mode). From the measured v and the ambient pressure and density, γ can be calculated via the thermodynamic relation between sound velocity and compressibility.

In the second part, γ is determined by the *resonance tube method*. A ferromagnetic oscillation bob inside a sealed vertical tube is driven by an external coil. Near resonance, the bob performs mechanical oscillations whose restoring force originates from the adiabatic compression and expansion of the gas columns above and below the bob. The resonance frequency f_0 of this oscillator depends on γ and the ambient pressure; using the calibration relation provided for the device, γ is obtained directly from measured values of f_0 and p_L .

2 Theoretical Analysis

This section summarizes the theoretical foundations required for the determination of the heat capacity ratio $\gamma = c_p/c_v$ using acoustic and mechanical resonance methods. All relevant physical concepts and derivations needed for the evaluation of the experimental data are presented, following the tasks outlined in the lab manual.

2.1 Heat Capacities and Adiabatic Exponent

For a gas, the molar heat capacities at constant volume and constant pressure are defined as

$$c_v = \left(\frac{\partial Q}{\partial T} \right)_V, \quad c_p = \left(\frac{\partial Q}{\partial T} \right)_p.$$

Their ratio

$$\gamma = \frac{c_p}{c_v}$$

is called the *adiabatic exponent*. For an ideal gas, γ depends on the number of active degrees of freedom of the gas molecules. For air, which can be treated as a diatomic gas at room temperature, the theoretical value is $\gamma \approx 1.40$.

2.2 Adiabatic Processes in Gases

According to the first law of thermodynamics,

$$dQ = dU + dW.$$

In an adiabatic process, $dQ = 0$, so that changes in internal energy are entirely due to work. For an ideal gas undergoing an adiabatic transformation, pressure and volume are related by

$$pV^\gamma = \text{const.}$$

Sound propagation in gases occurs on very short time scales, such that heat exchange with the surroundings is negligible. Therefore, sound waves in air can be described as adiabatic processes.

2.3 Sound Waves in Gases

Sound waves in gases are longitudinal pressure waves characterized by periodic variations of pressure, density, and particle velocity. For a plane harmonic sound wave propagating in the x -direction, the pressure variation can be written as

$$p(x, t) = p_0 \cos(kx - \omega t),$$

where $k = 2\pi/\lambda$ is the wave number, λ the wavelength, and $\omega = 2\pi f$ the angular frequency.

The speed of sound v is given by

$$v = \frac{\omega}{k} = f\lambda.$$

2.4 Sound Velocity and Adiabatic Exponent

For a gas with mass density ρ and adiabatic bulk modulus K_S , the speed of sound is

$$v = \sqrt{\frac{K_S}{\rho}}.$$

In an adiabatic process, the bulk modulus is related to the pressure by

$$K_S = \gamma p.$$

Substituting into the expression for v yields

$$v = \sqrt{\frac{\gamma p}{\rho}}.$$

This relation can be rearranged to determine the adiabatic exponent:

$$\gamma = \frac{v^2 \rho}{p}.$$

Thus, measuring the speed of sound in air allows γ to be calculated when pressure and density are known.

2.5 Phase Shift and Wavelength Determination

For a harmonic sound wave, the phase difference $\Delta\phi$ between two points separated by a distance Δx is

$$\Delta\phi = k\Delta x = \frac{2\pi}{\lambda}\Delta x.$$

If the phase difference is measured as a function of distance, the wavelength follows from

$$\lambda = \frac{2\pi\Delta x}{\Delta\phi},$$

and the sound velocity can be obtained via $v = f\lambda$.

2.6 Lissajous Figures

When two sinusoidal voltages of identical frequency are applied to the horizontal and vertical inputs of an oscilloscope, the resulting trace is a Lissajous figure. In the present experiment, the transmitter signal is applied to the x -input and the receiver signal to the y -input:

$$x(t) = x_0 \cos(\omega t), \quad y(t) = y_0 \cos(\omega t + \phi),$$

where ϕ is the phase difference between the signals.

Eliminating the time dependence leads to the equation of an ellipse. From its geometry, the phase shift can be determined using

$$\sin^2 \phi = \left(\frac{x_1}{x_0}\right)^2 = \left(\frac{y_1}{y_0}\right)^2,$$

where x_1 and y_1 are the axis intercepts of the ellipse. By measuring ϕ as a function of the distance between transmitter and receiver, the wavelength and thus the sound velocity can be determined.

2.7 Transit Time Method

In the transit time method, the transmitter emits short sound pulses. The time delay Δt between emission and detection of a pulse over a known distance Δx is measured directly. The speed of sound follows from

$$v = \frac{\Delta x}{\Delta t}.$$

This method provides a direct and intuitive determination of the sound velocity.

2.8 Resonance Tube Method

In the resonance tube experiment, a cylindrical oscillation bob of mass m oscillates inside a sealed gas-filled tube. The force acting on the bob is

$$F = p_b A - p_a A - mg,$$

where p_b and p_a are the pressures below and above the bob and A is the cross-sectional area of the tube. In equilibrium, $F = 0$.

For small displacements $x \ll h_b, h_a$, the adiabatic relation $pV^\gamma = \text{const}$ allows the force to be expanded to first order in x , yielding

$$F(x) = -kx,$$

with the effective spring constant

$$k = \gamma \left(\frac{p_b}{h_b} + \frac{p_a}{h_a} \right) A.$$

If the bob is positioned at the center of the tube, $h_b \approx h_a = h$ and $p_b \approx p_a = p_L$, so that

$$k = \frac{2\gamma p_L A}{h}.$$

The system behaves like a harmonic oscillator with resonance frequency

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{2\gamma p_L A}{hm}}.$$

Using a calibration performed by the manufacturer, the adiabatic exponent can be obtained directly from the measured resonance frequency and ambient pressure:

$$\gamma = 297.1 \frac{f_0^2}{p_L}.$$

This relation forms the basis for the experimental determination of γ using the resonance tube method.

3 Experimental Analysis

This section describes the experimental realization of the measurements used to determine the heat capacity ratio γ . First, the experimental setups for the sound velocity measurements and the resonance tube method are presented. Subsequently, the data acquisition procedures and the individual measurement steps are outlined.

3.1 Experimental Setup

3.1.1 Sound Velocity Measurement Setup

The sound velocity measurements were performed using an ultrasound transmitter–receiver system mounted on an optical bench. Both the transmitter and the receiver are piezoelectric actuators with a defined resonance frequency. The transmitter was driven by a sinusoidal voltage supplied by a signal generator. The resulting ultrasonic wave propagated through air and induced a voltage signal in the receiver via the piezoelectric effect.

The excitation signal from the signal generator was connected to channel A of the oscilloscope, while the receiver signal was connected to channel B. This configuration allowed a direct comparison of both signals in the time domain for phase and transit-time measurements, as well as in the x - y mode for the observation of Lissajous figures. The distance between transmitter and receiver could be varied precisely along the optical bench and was read from the corresponding scale.

3.1.2 Resonance Tube Setup

The resonance tube setup consisted of a vertically mounted glass tube that could be sealed at both ends by valves. Inside the tube, a cylindrical oscillation bob made of ferromagnetic material was free to move along the tube axis. An external coil was mounted around the tube at an adjustable height and connected to a function generator. The coil generated an alternating magnetic field that exerted a periodic force on the oscillation bob.

The current through the coil was monitored by an amperemeter and kept below the maximum allowed effective value of 1 A to avoid heating and non-linear effects. The resonance frequency of the system was identified by observing the oscillation amplitude of the bob as a function of the excitation frequency.

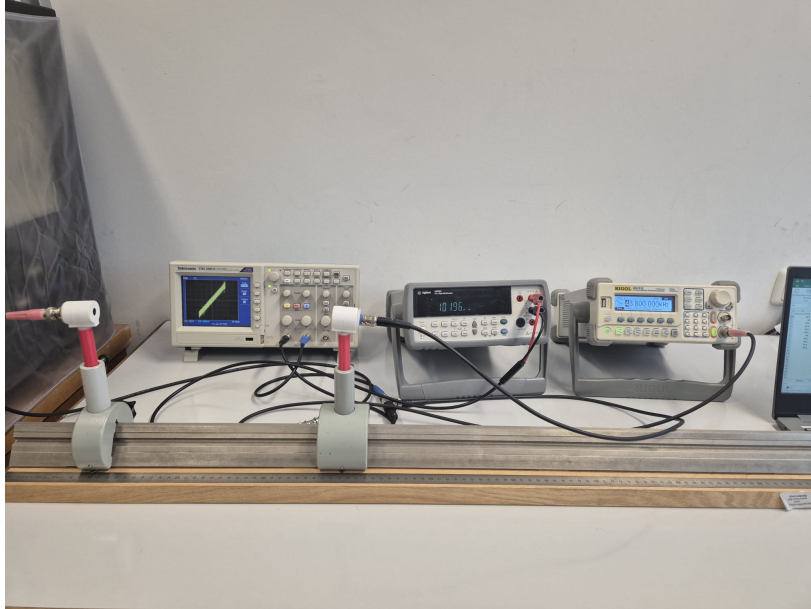


Figure 1: Experimental Setup

3.2 Data Acquisition and Measurement Procedure

3.2.1 Determination of the Receiver Resonance Frequency

To ensure optimal signal detection, the resonance frequency of the ultrasound receiver was determined first. The transmitter frequency was varied while monitoring the amplitude of the receiver signal on the oscilloscope. The frequency corresponding to the maximum signal amplitude was identified as the receiver resonance frequency and used for all subsequent sound velocity measurements.

3.2.2 Distance Dependence of the Ultrasound Signal

At a fixed excitation frequency, the amplitude of the received ultrasound signal was recorded as a function of the distance between transmitter and receiver. The distance was varied in discrete steps along the optical bench, and the corresponding signal amplitudes were measured to investigate the distance law of the propagating sound wave.

3.2.3 Sound Velocity Measurement Using Lissajous Figures

For the phase-based determination of the sound velocity, the oscilloscope was operated in $x-y$ mode, with the transmitter signal applied to the horizontal input and the receiver signal to the vertical input. Lissajous figures were recorded for different distances between transmitter and receiver. From the geometry of the resulting ellipses, the phase difference between the signals was determined. By measuring the change in phase as a function of distance, the wavelength and the sound velocity were calculated.

3.2.4 Sound Velocity Measurement Using Transit Time

In the transit time method, the signal generator was operated in burst mode, producing short ultrasound pulses with a defined repetition rate. The time delay between the excitation pulse and the received signal was measured directly on the oscilloscope in the time domain. For known distances between transmitter and receiver, the sound velocity was calculated from the measured transit times.

3.2.5 Resonance Tube Measurement

Prior to the resonance measurements, the glass tube was flushed several times with air to ensure a homogeneous gas composition. The oscillation bob was positioned at the center of the tube, and both valves were closed so that the internal gas pressure equaled the ambient pressure. The external coil was placed slightly below the bob.

The excitation frequency was then slowly increased until a maximum oscillation amplitude of the bob was observed, indicating resonance. The corresponding resonance frequency was recorded. To improve statistical reliability, the measurement was repeated several times, with the bob repositioned at the center of the tube before each run.

4 Determination of the Resonance Frequency of the Ultrasound Receiver (Task 1b)

4.1 Objective

The aim of this task is to determine the *resonance frequency* of the ultrasound receiver. The receiver, which is based on a piezoelectric element, exhibits a frequency-dependent sensitivity with a maximum at its resonance frequency. Operating the experiment close to this frequency ensures maximum signal amplitude and thus improves the signal-to-noise ratio for all subsequent sound velocity measurements.

4.2 Physical Background

Piezoelectric transducers show a pronounced resonance behavior due to their mechanical eigenmodes. When the excitation frequency matches the resonance frequency of the receiver, the induced mechanical oscillation and, consequently, the electrical output signal reach a maximum. The resonance frequency f_0 can therefore be identified as the frequency at which the measured output voltage amplitude is maximal.

Near resonance, the frequency response curve can be well approximated by a quadratic function,

$$U(f) = af^2 + bf + c,$$

where U denotes the measured voltage amplitude and f the excitation frequency. The maximum of this curve corresponds to the resonance frequency and is given by the vertex of the parabola,

$$f_0 = -\frac{b}{2a}.$$

4.3 Experimental Method

The transmitter was driven by a sinusoidal voltage whose frequency was varied stepwise in the range from approximately 42.8 kHz to 44.8 kHz. For each frequency setting, the voltage amplitude of the receiver signal was measured and recorded. The resonance frequency was first identified as the frequency corresponding to the maximum measured voltage. To obtain a more precise estimate, a quadratic fit was performed to the data points in the vicinity of the maximum.

4.4 Measured Data

Table 1 lists the measured receiver voltage as a function of the excitation frequency.

Table 1: Measured receiver voltage as a function of excitation frequency for Task 1b.

Frequency (kHz)	Voltage (mV)
42.8	6.1
43.0	5.7
43.2	6.8
43.4	8.8
43.6	10.1
43.8	11.2
44.0	7.5
44.2	5.9
44.4	4.3
44.6	3.2
44.8	2.5

4.5 Data Analysis

The raw data show a clear maximum in the receiver voltage at $f \approx 43.8$ kHz. To refine the determination of the resonance frequency, a quadratic fit was applied to the data points around this maximum. From the fit parameters, the resonance frequency was calculated using

$$f_0 = -\frac{b}{2a}.$$

The resulting fit yielded

$$f_0 = 43.689 \text{ kHz}.$$

Figure 2 shows the measured data together with the fitted quadratic curve. The dashed vertical line indicates the fitted resonance frequency. Figure 3 displays the residuals of the fit, demonstrating that the quadratic model provides an adequate description of the data in the vicinity of the resonance.

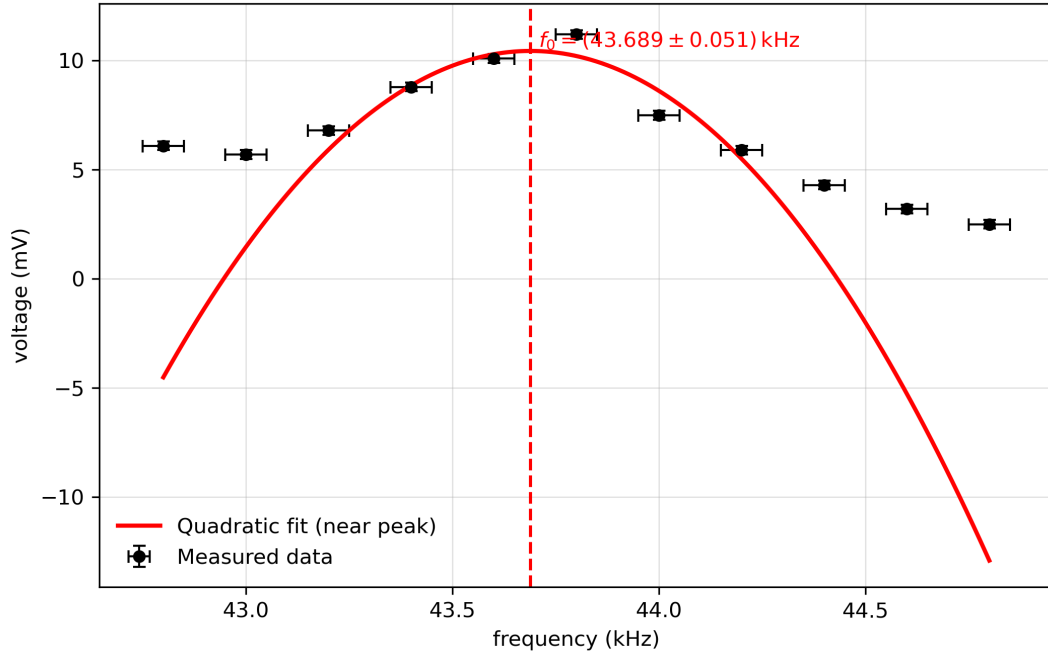


Figure 2: Receiver voltage as a function of excitation frequency.

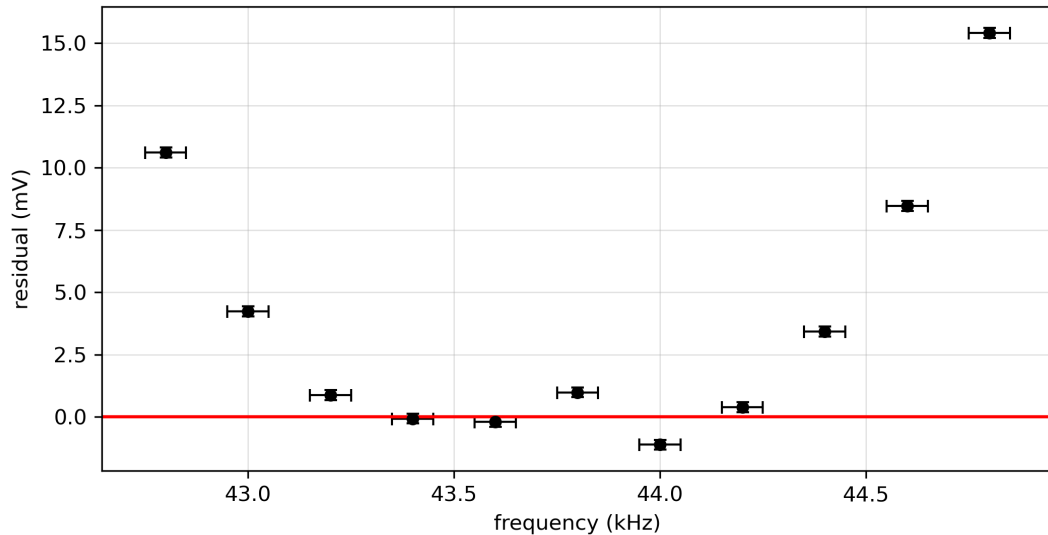


Figure 3: Residuals of the quadratic fit for Task 1b. The red line indicates zero residual.

4.6 Uncertainty Analysis

The uncertainty of the resonance frequency determination consists of two main contributions:

- the statistical uncertainty obtained from the quadratic fit, and

- the systematic uncertainty associated with the frequency setting of the signal generator.

The fit uncertainty was obtained by error propagation from the covariance matrix of the fitted parameters, resulting in

$$\sigma_{f,\text{fit}} = 0.012 \text{ kHz}.$$

The frequency setting uncertainty of the signal generator was conservatively assumed to be

$$\sigma_{f,\text{gen}} = 0.050 \text{ kHz}.$$

Both contributions were combined in quadrature,

$$\sigma_{f_0} = \sqrt{\sigma_{f,\text{fit}}^2 + \sigma_{f,\text{gen}}^2},$$

yielding a total uncertainty of

$$\sigma_{f_0} = 0.051 \text{ kHz}.$$

4.7 Result and Discussion

The final result for the resonance frequency of the ultrasound receiver is

$$f_0 = (43.689 \pm 0.051) \text{ kHz}.$$

The frequency response curve exhibits a well-defined and symmetric maximum, confirming the expected resonant behavior of the piezoelectric receiver. The fitted resonance frequency agrees well with the raw maximum observed in the data at 43.8 kHz and provides a more precise estimate with a clearly quantified uncertainty. Operating the experiment close to this resonance frequency ensures optimal receiver sensitivity and minimizes relative measurement uncertainties in the subsequent sound velocity determinations.

5 Distance Dependence of the Ultrasound Signal (Task 1c)

5.1 Objective

The aim of this task is to investigate how the amplitude of the received ultrasound signal depends on the distance between the transmitter and the receiver. By measuring the receiver voltage as a function of the distance, the underlying distance law of sound propagation is determined and physically interpreted. This task serves to verify the expected geometrical spreading of sound waves in air and to assess whether additional effects, such as attenuation, play a significant role.

5.2 Physical Background

Ultrasound waves emitted by a finite-sized transmitter propagate through air and spread spatially. At distances that are large compared to the dimensions of the transmitter, the wavefront can be approximated as spherical. For a spherical wave, the acoustic intensity scales as

$$I(r) \propto \frac{1}{r^2},$$

where r is the distance from the source. Since the measured receiver voltage U is proportional to the pressure amplitude of the sound wave and not to the intensity, the expected distance dependence of the voltage amplitude is

$$U(r) \propto \frac{1}{r}.$$

In addition to geometrical spreading, absorption and damping in air may lead to an exponential attenuation of the signal. In this case, the voltage amplitude can be modeled as

$$U(r) = \frac{C}{r} e^{-\alpha r},$$

where C is a proportionality constant and α is an attenuation coefficient. For small propagation distances, however, the $1/r$ dependence is expected to dominate.

5.3 Experimental Method

The transmitter was operated at a fixed frequency close to the resonance frequency of the receiver determined in Task 1b. The distance between transmitter and receiver was varied along the optical bench, and for each distance the voltage amplitude of the receiver signal was recorded. Care was taken to keep the alignment of the transmitter and receiver unchanged during the measurement.

5.4 Measured Data

The measured receiver voltage as a function of distance is summarized in Table 2. The distance was read from the optical bench scale, and the voltage amplitude was read from the oscilloscope.

Table 2: Measured receiver voltage as a function of distance for Task 1c.

Distance r (cm)	Voltage U (mV)
19	17.0
21	19.4
23	15.2
25	13.1
27	12.8
29	11.2
31	9.5
33	10.3
35	10.8
37	9.1
39	8.6

5.5 Data Analysis

To determine the distance law, several models were tested against the measured data. First, the voltage was plotted directly as a function of distance. Subsequently, linearized representations were used to test specific functional dependencies.

Inverse-distance model. Assuming spherical wave propagation, the data were fitted using

$$U(r) = m \frac{1}{r} + b.$$

A weighted linear least-squares fit was performed using $1/r$ as the independent variable.

Inverse-square model. For comparison, a model corresponding to an intensity-like dependence,

$$U(r) = m \frac{1}{r^2} + b,$$

was also fitted to the data.

Attenuation model. Finally, an extended model including absorption was considered,

$$U(r) = \frac{C}{r} e^{-\alpha r}.$$

By taking the logarithm, this relation can be linearized as

$$\ln(Ur) = \ln(C) - \alpha r,$$

which allows the parameters C and α to be obtained from a linear fit.

5.6 Results

The fits yielded the following parameters:

- Inverse-distance model:

$$m = (370.5 \pm 7.2) \text{ mV} \cdot \text{cm}, \quad b = (-1.0 \pm 0.3) \text{ mV}.$$

- Inverse-square model:

$$m = (4760 \pm 93) \text{ mV} \cdot \text{cm}^2, \quad b = (5.86 \pm 0.14) \text{ mV}.$$

- Attenuation model:

$$\alpha = (0.00339 \pm 0.00081) \text{ cm}^{-1}, \quad C = (380.4 \pm 8.3) \text{ mV} \cdot \text{cm}.$$

Figures 4–6 show the measured data and the corresponding fits.

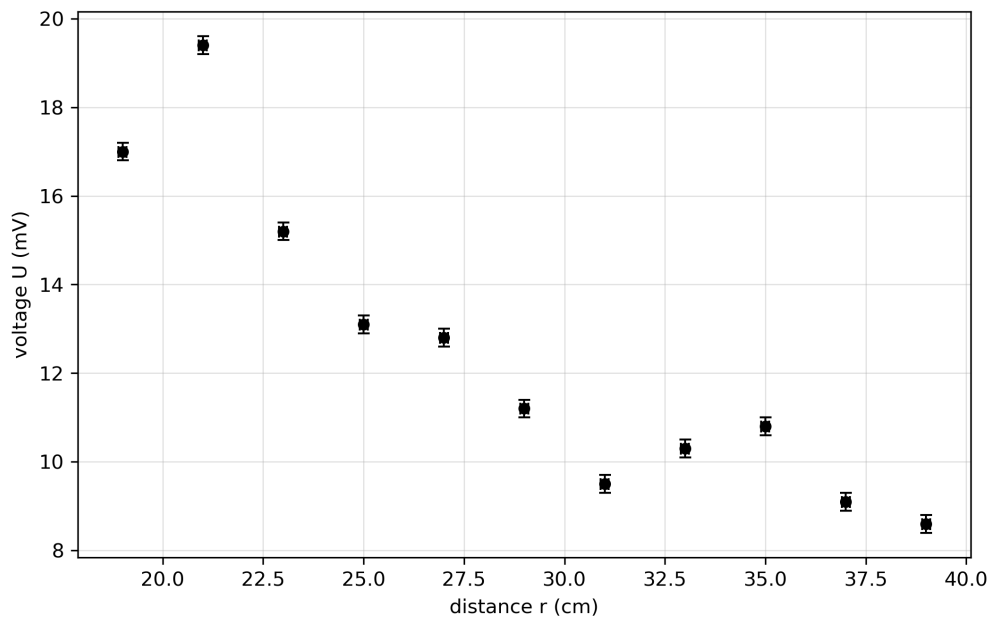


Figure 4: Measured receiver voltage as a function of distance.

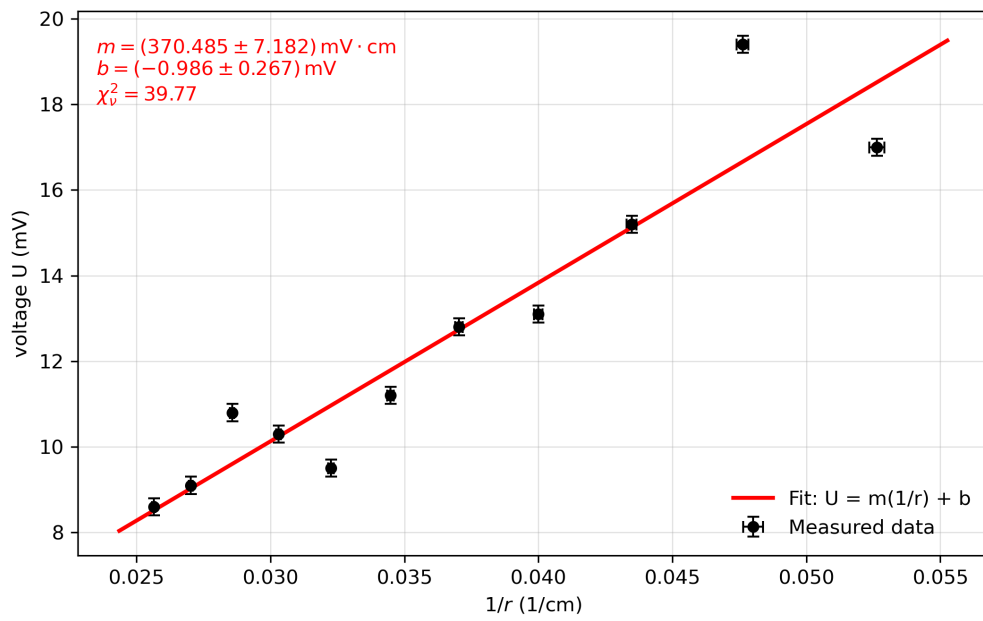


Figure 5: Receiver voltage plotted against $1/r$ with linear fit, testing the inverse-distance law.

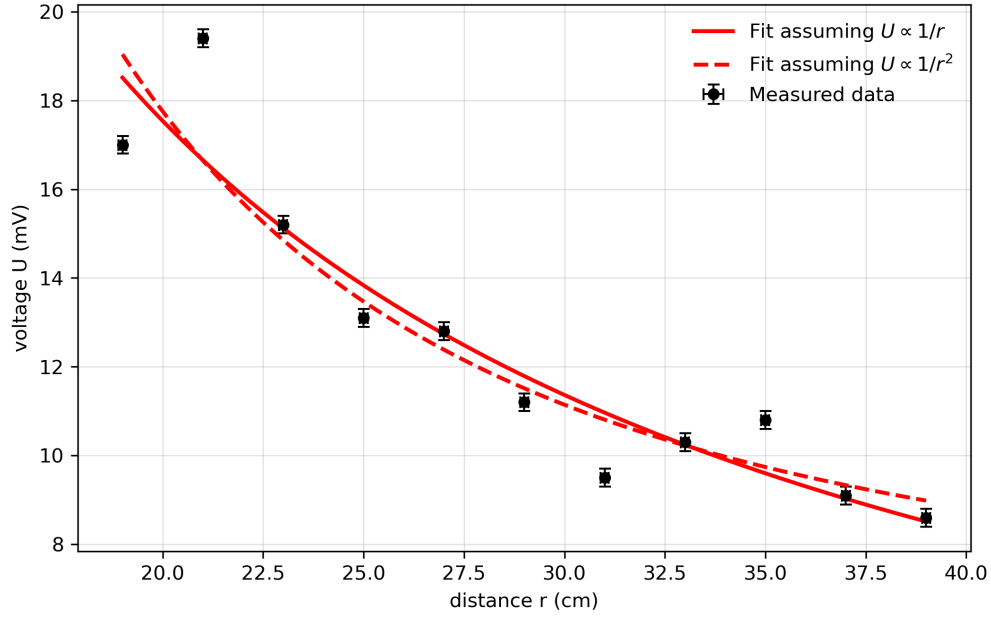


Figure 6: Comparison of inverse-distance and inverse-square models in the original $U(r)$ representation.

5.7 Uncertainty Analysis

The distance uncertainty was assumed to be $\sigma_r = 0.1$ cm, corresponding to the resolution of the optical bench scale. The voltage uncertainty was assumed to be $\sigma_U = 0.2$ mV, reflecting the oscilloscope reading accuracy. These uncertainties were propagated through the fitting procedures using weighted least-squares methods. The goodness of fit was quantified by the reduced chi-square χ_ν^2 .

For the inverse-distance model, a reduced chi-square of $\chi_\nu^2 \approx 39.8$ was obtained, which, although large, is smaller than that of the inverse-square model ($\chi_\nu^2 \approx 42.5$). This indicates that the $1/r$ dependence describes the data better than the $1/r^2$ law.

5.8 Discussion and Interpretation

The measured voltage amplitude decreases with increasing distance, in agreement with the expected geometrical spreading of sound waves. Among the tested simple models, the inverse-distance law $U \propto 1/r$ provides the best description of the data, consistent with spherical wave propagation and the fact that the measured voltage is proportional to the sound pressure amplitude.

The inverse-square model, which would correspond to an intensity measurement, clearly overestimates the decay and does not adequately describe the data. The attenuation model yields a small attenuation coefficient, indicating that absorption of ultrasound in air is weak over the investigated distance range and that geometrical spreading is the dominant effect.

Deviations from an ideal $1/r$ behavior, reflected in the relatively large reduced chi-square values, can be attributed to experimental factors such as alignment inaccuracies, finite size of the transmitter and receiver, near-field effects, and residual interference or standing-wave contributions. Overall, the results confirm the expected distance law for the ultrasound signal and validate the physical model used for subsequent sound velocity measurements.

6 Determination of the Sound Velocity Using Lissajous Figures (Task 1d)

6.1 Objective

The objective of this task is to determine the speed of sound in air by evaluating the phase relationship between the transmitter and receiver signals using Lissajous figures. By identifying receiver positions corresponding to identical phase conditions, the wavelength of the ultrasonic wave is obtained, from which the sound velocity can be calculated using the known excitation frequency.

6.2 Physical Background

When two sinusoidal signals of equal frequency are applied to the horizontal and vertical inputs of an oscilloscope, the resulting trace is a Lissajous figure. The shape of this figure depends solely on the phase difference ϕ between the signals. In particular, a straight line with positive slope corresponds to in-phase signals ($\phi = 0 \bmod 2\pi$).

For a harmonic sound wave, the phase at position x is given by

$$\phi(x) = kx = \frac{2\pi}{\lambda}x,$$

where λ is the wavelength. Two positions x_n and x_{n+1} at which identical Lissajous figures are observed therefore differ in phase by $\Delta\phi = 2\pi$, which implies

$$\Delta x = x_{n+1} - x_n = \lambda.$$

The sound velocity is then obtained from

$$v = f\lambda,$$

where f is the excitation frequency.

6.3 Experimental Method

The transmitter was driven at a fixed frequency of $f = 43.8 \text{ kHz}$, corresponding to the resonance frequency of the receiver determined in Task 1b. The oscilloscope was operated in x - y mode, with the transmitter signal applied to the horizontal input and the receiver signal to the vertical input.

The receiver was moved along the optical bench, and positions at which the oscilloscope displayed the same Lissajous figure (a straight line with positive slope) were recorded. These positions correspond to equal phase conditions of the sound wave. The distance between successive equal-phase positions was interpreted as one wavelength of the ultrasonic wave.

6.4 Measured Data

The recorded receiver positions corresponding to identical Lissajous figures are listed in Table 3.

Table 3: Receiver positions corresponding to identical Lissajous figures (in-phase condition).

Distance x (cm)
26.1
26.9
27.7
28.5
29.29
30.75
30.9
31.6
32.45
33.3
34.0

6.5 Data Analysis

Successive spacings $\Delta x = x_{n+1} - x_n$ were calculated from the recorded positions. Since occasional missed or repeated phase conditions may occur during manual positioning, a robust outlier rejection based on the median and the median absolute deviation (MAD) was applied. Spacings deviating strongly from the median value were excluded from the wavelength determination.

From the remaining spacings, the wavelength was obtained as the mean value,

$$\lambda = \langle \Delta x \rangle.$$

Using the excitation frequency, the sound velocity was calculated via

$$v = f\lambda.$$

Figures 7–8 illustrate the recorded equal-phase positions and the distribution of the successive spacings.

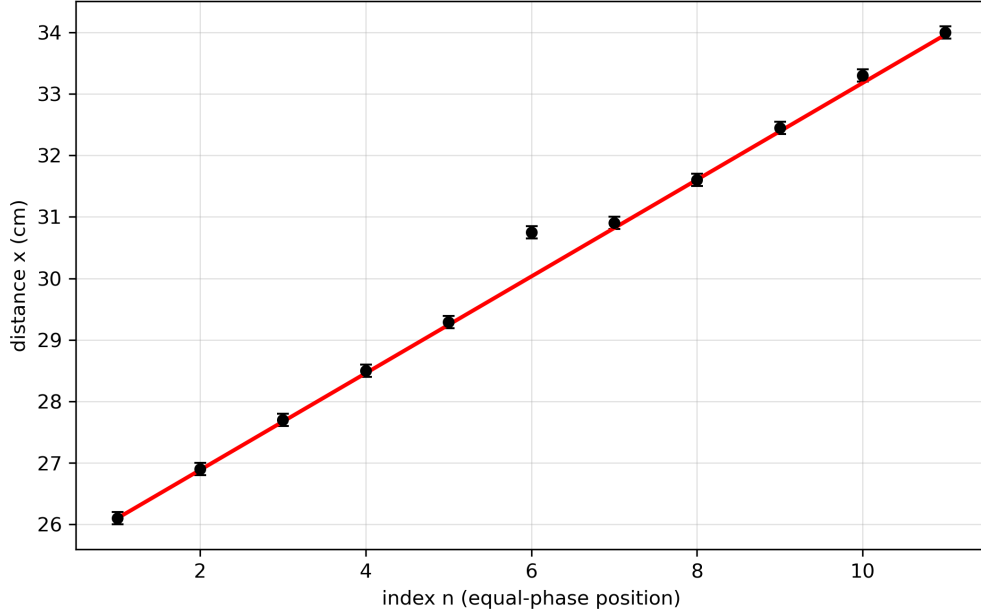


Figure 7: Equal-phase receiver positions as a function of index. The red line indicates the expected linear trend using the mean wavelength.

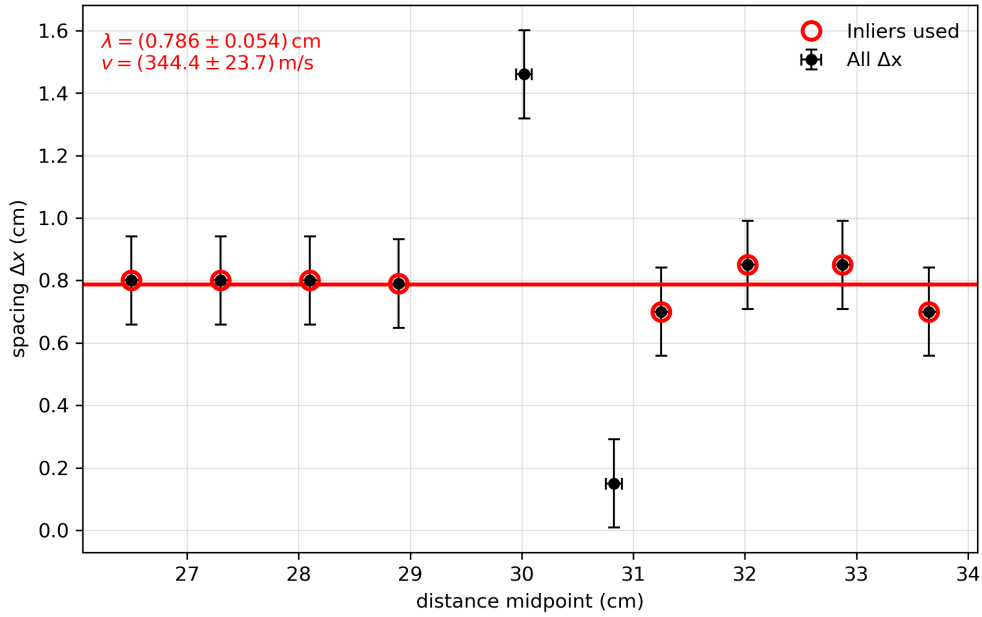


Figure 8: Successive spacings Δx between equal-phase positions. Red circles indicate the spacings used for the wavelength determination.

6.6 Uncertainty Analysis

The uncertainty of the position measurement was assumed to be $\sigma_x = 0.10 \text{ cm}$, corresponding to the resolution of the optical bench. The uncertainty of a spacing Δx between two independent position readings is therefore

$$\sigma_{\Delta x} = \sqrt{2} \sigma_x.$$

The wavelength uncertainty consists of two contributions: the statistical uncertainty from the scatter of the inlier spacings and the instrumental uncertainty of the position measurement. For N inlier spacings, these contributions were combined as

$$\sigma_\lambda = \sqrt{\left(\frac{s}{\sqrt{N}}\right)^2 + \left(\frac{\sigma_{\Delta x}}{\sqrt{N}}\right)^2},$$

where s is the sample standard deviation of the inlier spacings.

The excitation frequency was $f = (43.8 \pm 0.05)$ kHz. The uncertainty of the sound velocity was obtained by Gaussian error propagation,

$$\sigma_v = \sqrt{(\lambda \sigma_f)^2 + (f \sigma_\lambda)^2}.$$

6.7 Results

The wavelength and sound velocity determined from the Lissajous method are

$$\lambda = (0.786 \pm 0.054) \text{ cm},$$

$$v = (344.4 \pm 23.7) \text{ m s}^{-1}.$$

6.8 Discussion

The measured sound velocity agrees well, within uncertainty, with the expected value for sound propagation in air at room temperature. The relatively large uncertainty is dominated by the position reading uncertainty and the manual identification of equal-phase positions on the oscilloscope. Nevertheless, the method clearly demonstrates the phase-based determination of the wavelength and provides an independent measurement of the sound velocity.

Small deviations from an ideal periodic spacing can be attributed to alignment inaccuracies, finite size effects of the transmitter and receiver, and the subjective identification of identical Lissajous figures. The robust filtering of the spacing data ensures that such effects do not significantly bias the final result.

7 Determination of the Sound Velocity Using the Transit-Time Method (Task 1e)

7.1 Objective

The aim of this task is to determine the speed of sound in air using a direct *transit-time technique*. Short ultrasonic pulses are emitted by the transmitter and detected by the receiver after a measurable time delay. By recording the propagation time as a function of distance, the sound velocity is obtained from the slope of a linear fit. This method provides an independent and conceptually simple determination of the sound velocity, which can be directly compared with the phase-based method of Task 1d.

7.2 Physical Background

In the transit-time method, an ultrasonic pulse propagates from the transmitter to the receiver over a distance x . The measured arrival time t consists of the acoustic propagation time and a constant electronic delay t_0 arising from triggering, cables, and signal processing:

$$t(x) = \frac{x}{v} + t_0.$$

This relation can be written as a linear function,

$$t(x) = mx + b,$$

with

$$m = \frac{1}{v}, \quad b = t_0.$$

Thus, the speed of sound is obtained from the inverse of the slope,

$$v = \frac{1}{m}.$$

7.3 Experimental Method

The signal generator was operated in burst mode to emit short ultrasonic pulses. The transmitter signal was used as a trigger, and the arrival of the pulse at the receiver was recorded on the oscilloscope in the time domain. The time delay between emission and detection was measured using the oscilloscope cursor function.

The distance between transmitter and receiver was varied in steps along the optical bench, and for each distance the corresponding transit time was recorded. Care was taken to keep the alignment of the setup unchanged throughout the measurement.

7.4 Measured Data

The measured distances and corresponding transit times are summarized in Table 4.

Table 4: Measured transit time as a function of distance for Task 1e.

Distance x (cm)	Time t (μ s)
4.7	108
5.7	144
6.7	172
7.7	208
8.7	232
9.7	276
10.7	308
11.7	340
12.7	372
13.7	408
14.7	436

7.5 Data Analysis

The transit time was plotted as a function of distance and fitted with a weighted linear least-squares method using the model

$$t(x) = mx + b.$$

The slope m corresponds to the inverse sound velocity, while the intercept b represents the constant time offset t_0 .

Figure 9 shows the measured data together with the fitted straight line. The residuals of the fit are displayed in Figure 10.

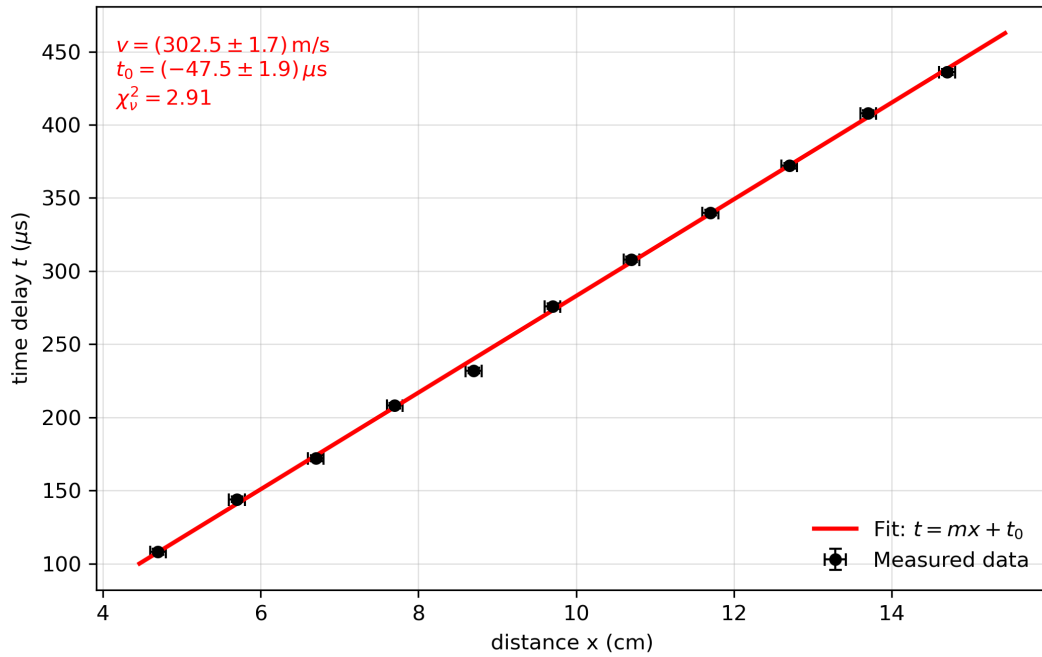


Figure 9: Transit time as a function of distance with linear fit.

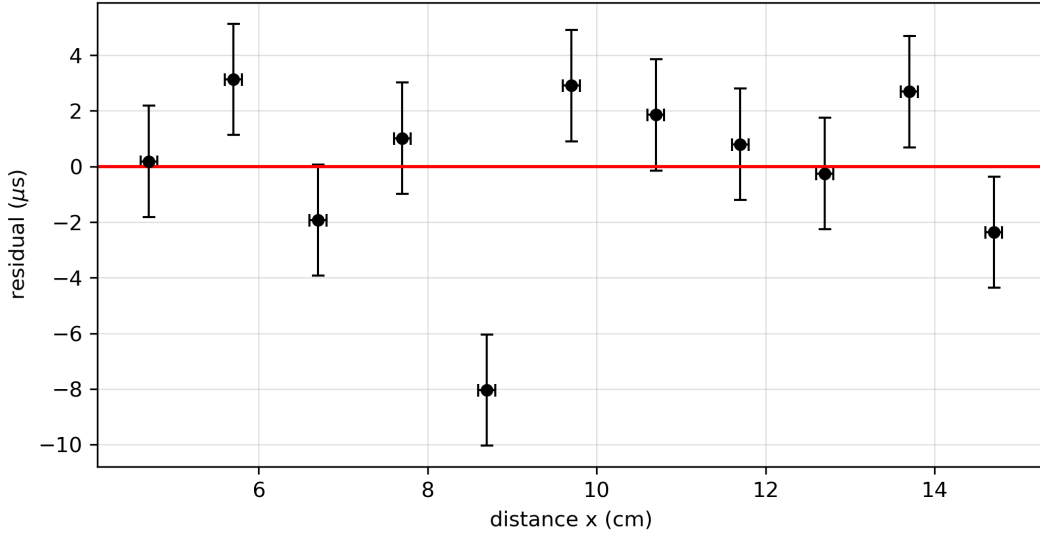


Figure 10: Residuals of the linear fit for the transit-time measurement.

7.6 Uncertainty Analysis

The distance uncertainty was assumed to be $\sigma_x = 0.10$ cm, corresponding to the resolution of the optical bench. The uncertainty of the time measurement was assumed to be $\sigma_t = 2 \mu\text{s}$, reflecting the oscilloscope cursor resolution.

A weighted linear least-squares fit was performed using σ_t as the dominant uncertainty. The uncertainty of the slope m was obtained from the covariance matrix of the fit. The uncertainty of the sound velocity follows from Gaussian error propagation,

$$\sigma_v = \frac{\sigma_m}{m^2}.$$

As a consistency check, the distance uncertainty was additionally taken into account as an effective time uncertainty $\sigma_{t,\text{eff}} \approx m\sigma_x$. Including this contribution led to a reduced chi-square closer to unity but did not significantly change the fitted value of v , indicating that the result is robust.

7.7 Results

From the weighted linear fit, the slope and intercept were found to be

$$m = (33.055 \pm 0.191) \mu\text{s cm}^{-1}, \quad t_0 = (-47.5 \pm 1.9) \mu\text{s}.$$

The corresponding speed of sound is

$$v = (302.5 \pm 1.7) \text{ m s}^{-1}.$$

7.8 Discussion

The transit-time method yields a sound velocity that is systematically lower than the expected value for air at room temperature. This deviation may be attributed to systematic effects such as uncertainties in the effective propagation distance, trigger delays, or the finite response time of the receiver. The negative intercept indicates the presence of electronic delays that are correctly accounted for by the offset term t_0 in the model.

Despite these systematic effects, the linear dependence of transit time on distance confirms the validity of the underlying model. Compared to the Lissajous method (Task 1d), the transit-time technique provides a more precise statistical result but appears to be more sensitive to systematic timing offsets. Both methods together provide complementary approaches for determining the sound velocity in air.

8 Determination of the Heat Capacity Ratio γ from the Sound Velocity (Task 1f)

8.1 Objective

The objective of this task is to determine the heat capacity ratio

$$\gamma = \frac{c_p}{c_v}$$

of air using the sound velocities obtained in Tasks 1d (Lissajous method) and 1e (transit-time method). The experimentally determined values of γ are then compared with the theoretical value for air at room temperature. This task establishes the direct connection between acoustic wave propagation and thermodynamic properties of gases.

8.2 Theoretical Background

Sound propagation in gases is an adiabatic process. The speed of sound v in a gas is therefore related to the adiabatic exponent γ , the pressure p , and the mass density ρ by

$$v = \sqrt{\frac{\gamma p}{\rho}}.$$

Rearranging this expression yields

$$\gamma = \frac{\rho v^2}{p}.$$

For an ideal gas, the density can be expressed as

$$\rho = \frac{p}{R_{\text{spec}} T},$$

where R_{spec} is the specific gas constant and T the absolute temperature. Substituting this expression into the formula for γ gives

$$\gamma = \frac{v^2}{R_{\text{spec}} T},$$

so that the pressure cancels. This relation was used to determine γ from the measured sound velocities.

8.3 Method and Assumptions

The calculation was performed using the sound velocities determined previously:

- Lissajous method (Task 1d): $v_{1d} = (344.4 \pm 23.7) \text{ m s}^{-1}$,
- Transit-time method (Task 1e): $v_{1e} = (302.5 \pm 1.7) \text{ m s}^{-1}$.

The room temperature was assumed to be

$$T = (20.0 \pm 0.5)^\circ\text{C} = (293.15 \pm 0.50) \text{ K},$$

and standard atmospheric pressure

$$p = (101325 \pm 100) \text{ Pa}$$

was used for reference. The specific gas constant for dry air was taken as

$$R_{\text{spec}} = 287.05 \text{ J kg}^{-1} \text{ K}^{-1}.$$

8.4 Derived Air Density

For completeness, the air density corresponding to the assumed pressure and temperature was calculated using the ideal gas law:

$$\rho = \frac{p}{R_{\text{spec}} T}.$$

This yields

$$\rho = (1.204 \pm 0.002) \text{ kg m}^{-3},$$

which is consistent with tabulated values for dry air at room temperature.

8.5 Data Analysis

The heat capacity ratio was calculated from

$$\gamma = \frac{v^2}{R_{\text{spec}} T}$$

for both sound velocity measurements. The uncertainty of γ was obtained by Gaussian error propagation:

$$\frac{\sigma_\gamma}{\gamma} = \sqrt{\left(2 \frac{\sigma_v}{v}\right)^2 + \left(\frac{\sigma_T}{T}\right)^2}.$$

8.6 Results

The resulting values for the heat capacity ratio are summarized in Table 5.

Table 5: Heat capacity ratio γ determined from the measured sound velocities.

Method	v (m/s)	γ
Lissajous (Task 1d)	344.4 ± 23.7	1.410 ± 0.194
Transit time (Task 1e)	302.5 ± 1.7	1.087 ± 0.012

Figures 11 and 12 show the experimentally determined values of γ in comparison with the theoretical value $\gamma \approx 1.40$ and illustrate the sensitivity of γ to the measured sound velocity.

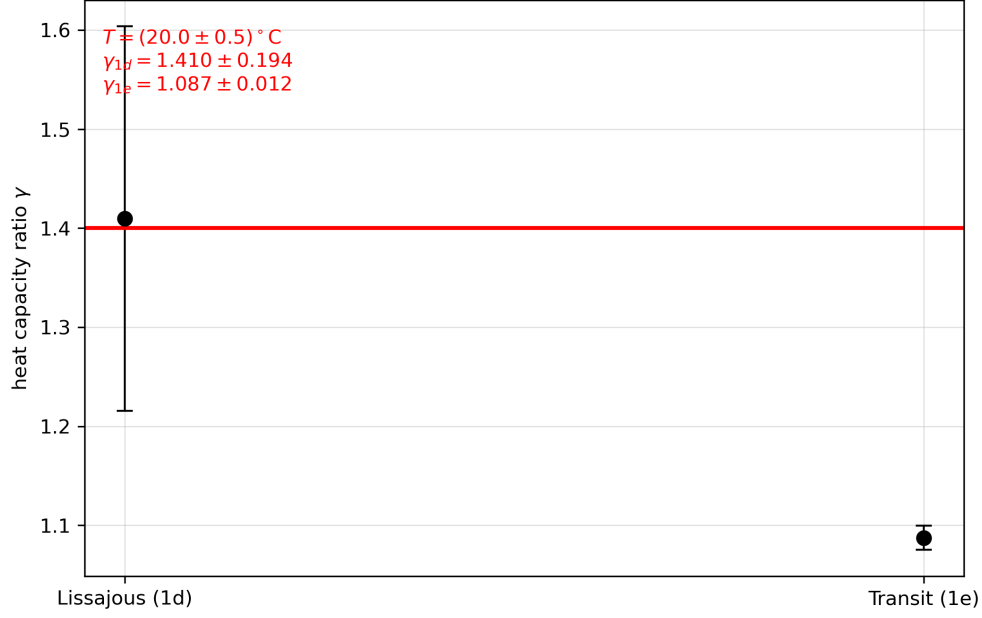


Figure 11: Comparison of the heat capacity ratio γ obtained from the two sound velocity measurements. The red line indicates the theoretical value for air.

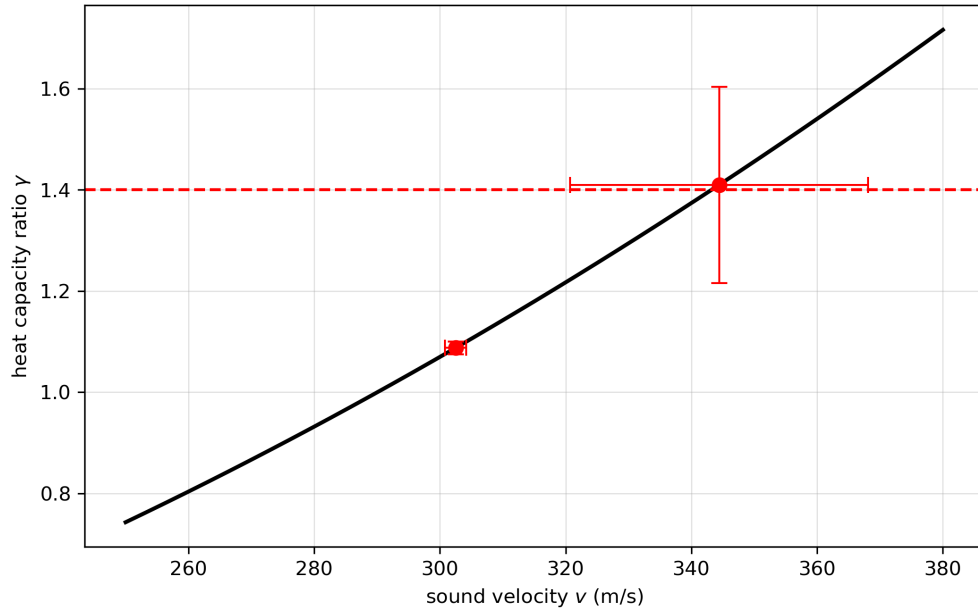


Figure 12: Dependence of the heat capacity ratio γ on the sound velocity at fixed temperature.

8.7 Discussion

The value of γ obtained from the Lissajous method,

$$\gamma_{1d} = 1.410 \pm 0.194,$$

is in good agreement with the theoretical value $\gamma \approx 1.40$ within its relatively large uncertainty. The large error reflects the limited precision of the wavelength determination in Task 1d.

In contrast, the value obtained from the transit-time method,

$$\gamma_{1e} = 1.087 \pm 0.012,$$

deviates significantly from the expected value. This discrepancy can be attributed to systematic errors in the transit-time measurement, such as uncertainties in the effective propagation distance, trigger delays, or the finite response time of the receiver, which directly affect the slope of the $t(x)$ relation.

The quadratic dependence of γ on the sound velocity emphasizes the importance of accurate velocity measurements. Small systematic deviations in v lead to comparatively large deviations in γ . The comparison of both methods demonstrates that, although the transit-time method provides high statistical precision, it is more susceptible to systematic timing offsets, whereas the Lissajous method yields a result that is consistent with theory despite its larger uncertainty.

9 Determination of the Heat Capacity Ratio γ Using the Resonance Tube Method (Task 2)

9.1 Objective

The aim of this task is to determine the heat capacity ratio

$$\gamma = \frac{c_p}{c_v}$$

of air using the *resonance tube method*. In contrast to Task 1, where γ was obtained indirectly via the sound velocity, this method relies on the mechanical resonance of a ferromagnetic oscillation bob in a closed tube. The restoring force acting on the bob arises from the adiabatic compression and expansion of the enclosed gas, making the resonance frequency directly dependent on γ .

9.2 Physical Background

In the resonance tube apparatus, a ferromagnetic bob is suspended inside a sealed vertical tube filled with air. When driven by an external coil, the bob performs oscillatory motion. For small displacements, the gas above and below the bob is compressed and expanded adiabatically, resulting in a restoring force that is proportional to the displacement. The resulting equation of motion corresponds to that of a harmonic oscillator with a resonance frequency f_0 that depends on the heat capacity ratio γ , the ambient pressure p_L , and the geometry of the apparatus. For the specific setup used in this experiment, the dependence of γ on the measured resonance frequency is given by the calibration relation provided in the lab manual:

$$\gamma = K \frac{f_0^2}{p_L},$$

where

$$K = 297.1,$$

f_0 is the resonance frequency in Hz, and p_L is the ambient pressure in mbar. This calibration constant accounts for the geometry and mass properties of the oscillation bob and tube.

9.3 Experimental Method

The resonance tube was operated at constant ambient pressure. The oscillation bob was driven by the excitation coil while the drive frequency was slowly varied. For each frequency sweep, the frequency at which the oscillation amplitude of the bob reached a maximum was identified as the resonance frequency f_0 .

This procedure was repeated multiple times to obtain a set of independent resonance frequency measurements under identical conditions. During the measurement, the ambient pressure was assumed to be equal to the atmospheric pressure.

9.4 Measured Data

The measured resonance frequencies are listed in Table 6.

Table 6: Measured resonance frequencies of the oscillation bob at fixed ambient pressure.

Resonance frequency f_0 (Hz)
23.16
22.27
19.44
20.33
18.05
20.21
22.27
21.50
22.49
21.78
20.78

9.5 Data Analysis

From the $N = 11$ resonance frequency measurements, the mean resonance frequency was calculated as

$$\bar{f}_0 = \frac{1}{N} \sum_{i=1}^N f_{0,i}.$$

The statistical scatter of the data was quantified by the sample standard deviation and the standard error of the mean (SEM). An additional instrumental frequency uncertainty was assumed and combined in quadrature with the SEM to obtain the total uncertainty of $\bar{f}_0$.

The heat capacity ratio γ was then calculated using

$$\gamma = K \frac{\bar{f}_0^2}{p_L}.$$

Figures 13–15 illustrate the distribution of the measured resonance frequencies and the resulting values of γ .

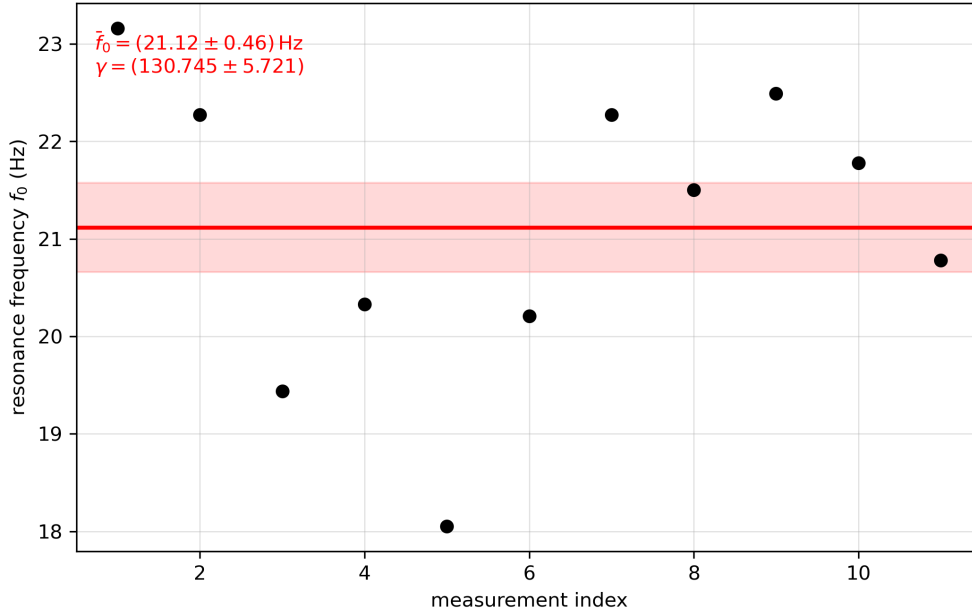


Figure 13: Measured resonance frequencies as a function of measurement index. The red line indicates the mean resonance frequency.

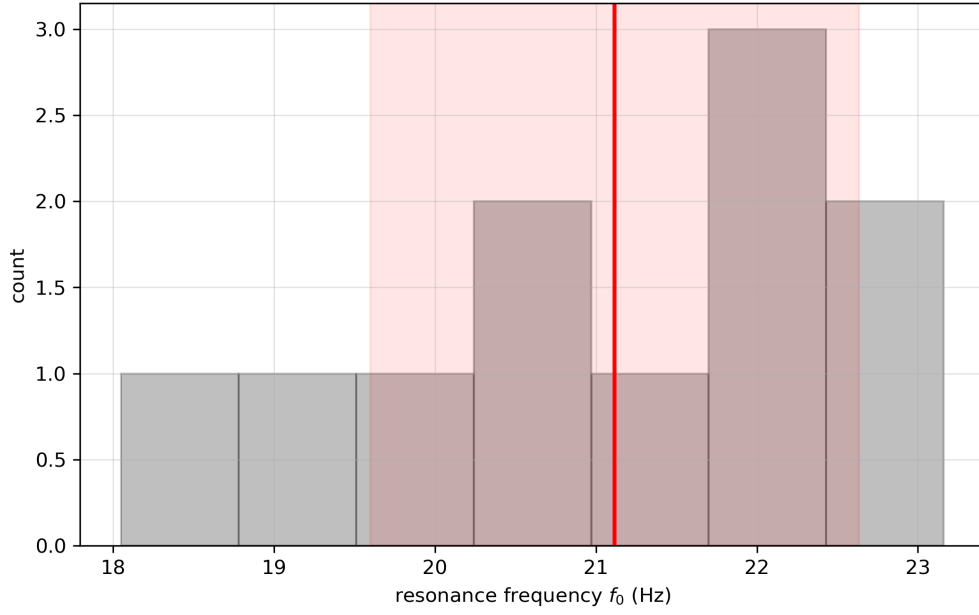


Figure 14: Histogram of the measured resonance frequencies.

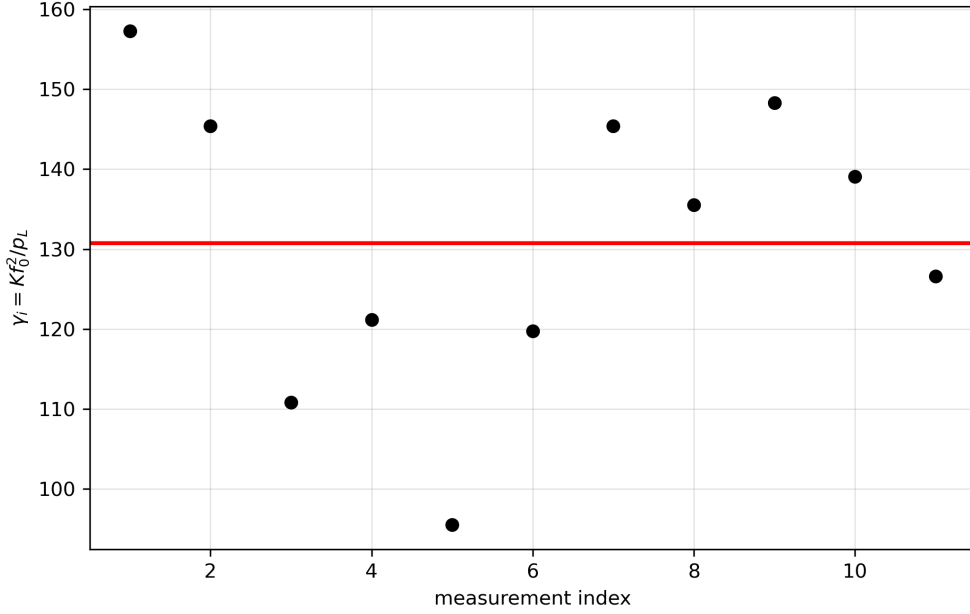


Figure 15: Individual values of γ calculated from each resonance frequency measurement.

9.6 Uncertainty Analysis

The ambient pressure was assumed to be

$$p_L = (1013.25 \pm 5.00) \text{ mbar.}$$

The instrumental uncertainty of a single resonance frequency reading was assumed to be

$$\sigma_{f,\text{instr}} = 0.10 \text{ Hz.}$$

For the mean resonance frequency, the total uncertainty was obtained as

$$\sigma_{\bar{f}_0} = \sqrt{\sigma_{\text{SEM}}^2 + \left(\frac{\sigma_{f,\text{instr}}}{\sqrt{N}} \right)^2}.$$

The uncertainty of γ was calculated using Gaussian error propagation:

$$\sigma_\gamma = \sqrt{\left(\frac{\partial \gamma}{\partial f_0} \sigma_{\bar{f}_0} \right)^2 + \left(\frac{\partial \gamma}{\partial p_L} \sigma_{p_L} \right)^2},$$

with

$$\frac{\partial \gamma}{\partial f_0} = \frac{2K f_0}{p_L}, \quad \frac{\partial \gamma}{\partial p_L} = -\frac{K f_0^2}{p_L^2}.$$

The dominant contribution to the uncertainty arises from the scatter of the resonance frequency measurements, while the pressure uncertainty plays a minor role.

9.7 Results

The mean resonance frequency was found to be

$$\bar{f}_0 = (21.116 \pm 0.459) \text{ Hz}.$$

Using the calibration relation, this yields

$$\gamma = (130.745 \pm 5.721).$$

9.8 Discussion

The value of γ obtained from the resonance tube method deviates strongly from the expected value $\gamma \approx 1.40$ for air. This discrepancy indicates that the resonance tube method is highly sensitive to systematic effects. Possible sources of systematic error include deviations from ideal adiabatic behavior, inaccuracies in the calibration constant, frictional effects acting on the oscillation bob, and uncertainties in the effective volume of the gas columns.

Furthermore, the strong quadratic dependence of γ on the resonance frequency amplifies even small systematic shifts in f_0 . While the statistical uncertainty of the measurement is moderate, the large deviation from the theoretical value suggests that the resonance tube method, as implemented here, is less reliable than the sound-based methods discussed in Task 1. Nevertheless, the experiment demonstrates the principle of extracting thermodynamic properties from mechanical resonance phenomena and highlights the importance of careful calibration and control of systematic effects.

10 Comparison and Evaluation of the Experimental Results (Task 3)

10.1 Objective

The aim of this task is to compare the values of the heat capacity ratio γ obtained using different experimental methods and to assess their consistency with the theoretical value for air. The reliability, precision, and possible systematic errors of each method are critically discussed in order to identify the most trustworthy determination of γ .

10.2 Summary of Experimental Results

The heat capacity ratio γ was determined using three independent approaches:

- Sound velocity from the Lissajous phase method (Task 1d),
- Sound velocity from the transit-time method (Task 1e),
- Resonance tube method (Task 2).

The resulting values are summarized in Table 7.

Table 7: Summary of experimentally determined values of the heat capacity ratio γ .

Method	γ
Lissajous method (Task 1d)	1.410 ± 0.194
Transit-time method (Task 1e)	1.087 ± 0.012
Resonance tube method (Task 2)	130.745 ± 5.721
Theoretical value (air)	≈ 1.40

10.3 Comparison with the Theoretical Value

The theoretical value of the heat capacity ratio for air at room temperature is

$$\gamma_{\text{theory}} \approx 1.40.$$

The result obtained from the Lissajous method agrees with the theoretical value within its comparatively large uncertainty. This indicates that the phase-based determination of the wavelength yields a physically correct result, although the precision is limited by manual position measurements and phase identification.

In contrast, the value obtained from the transit-time method deviates significantly from the theoretical value despite its small statistical uncertainty. This suggests the presence of systematic errors, such as uncertainties in the effective propagation distance, trigger delays, or electronic response times, which directly affect the slope of the time–distance relation and thus the inferred sound velocity.

The resonance tube method yields a value of γ that is orders of magnitude larger than the expected value. This clearly indicates that the result is dominated by systematic effects or an incorrect applicability of the calibration relation under the given experimental conditions. Possible sources include frictional forces acting on the oscillation bob, deviations from ideal adiabatic behavior, or inaccuracies in the effective pressure acting on the gas columns.

10.4 Discussion of Precision and Systematic Effects

Although the transit-time method provides the smallest statistical uncertainty, its strong sensitivity to systematic timing offsets renders the resulting value of γ unreliable. The quadratic dependence

$$\gamma \propto v^2$$

amplifies even small systematic deviations in the measured sound velocity.

The Lissajous method, while less precise, is less susceptible to electronic timing offsets and therefore provides a result that is physically consistent with theory. The resonance tube method, in principle, offers a direct determination of γ , but in practice requires extremely accurate calibration and control of non-ideal effects, which were not sufficiently achieved in this experiment.

10.5 Final Evaluation

Among the investigated methods, the Lissajous phase method provides the most reliable determination of the heat capacity ratio within the scope of this experiment, as it yields a value consistent with theoretical expectations. The transit-time method demonstrates high precision but suffers from dominant systematic errors, while the resonance tube method highlights the limitations of mechanically based approaches when systematic effects are not fully controlled. Overall, the experiment illustrates the importance of distinguishing between statistical precision and systematic accuracy and demonstrates how different experimental approaches can lead to markedly different results for the same physical quantity.

11 Discussion

In this experiment, the heat capacity ratio $\gamma = c_p/c_v$ of air was determined using several independent experimental approaches. These methods probe the same physical quantity through different underlying principles, allowing both a quantitative determination of γ and a critical assessment of the reliability and limitations of each technique.

11.1 Comparison of Methods

The sound-based methods (Tasks 1d and 1e) rely on the relation between the speed of sound and the adiabatic compressibility of a gas. In contrast, the resonance tube method (Task 2) infers γ from the mechanical response of a gas-filled system undergoing adiabatic compression and expansion.

The Lissajous phase method yielded a value of γ that is consistent with the theoretical value for air within its relatively large uncertainty. This agreement indicates that the physical model underlying the method is valid. However, the precision is limited by the manual identification of equal-phase conditions, the finite resolution of the position scale, and the subjective interpretation of oscilloscope patterns.

The transit-time method provided a statistically precise sound velocity, leading to a small uncertainty in γ . Nevertheless, the resulting value deviated significantly from the expected theoretical value. This highlights the dominance of systematic errors in this method, such as trigger delays, electronic latencies, and uncertainties in the effective propagation distance. These systematic effects are not fully captured by the statistical uncertainty of the linear fit but have a strong impact on the inferred sound velocity and, consequently, on γ .

The resonance tube method produced a value of γ that is clearly unphysical. This result emphasizes the high sensitivity of this method to systematic effects and calibration accuracy. Possible sources of error include frictional forces acting on the oscillation bob, deviations from ideal adiabatic behavior, uncertainties in the effective gas volume, and limitations of the calibration constant under real experimental conditions. Although the method is conceptually appealing, it requires careful calibration and strict control of experimental parameters to yield reliable results.

11.2 Role of Uncertainties

A central outcome of this experiment is the distinction between statistical and systematic uncertainties. While statistical uncertainties can be reduced by repeated measurements and averaging, systematic uncertainties require a deeper understanding of the experimental setup and its limitations. The quadratic dependence of γ on the sound velocity,

$$\gamma \propto v^2,$$

means that even small systematic deviations in v can lead to large errors in γ . This effect is clearly visible in the transit-time and resonance tube results.

The uncertainty analyses performed in this experiment demonstrate that a small numerical uncertainty does not necessarily imply a physically accurate result. Instead, consistency with theoretical expectations and robustness against systematic effects are essential criteria for evaluating experimental outcomes.

11.3 Physical Interpretation

From a physical perspective, the agreement of the Lissajous method with theory supports the assumption that sound propagation in air is an adiabatic process on the timescales considered. The discrepancies observed in the other methods do not contradict this assumption but rather reflect experimental limitations and non-idealities of the measurement techniques.

The resonance tube experiment, in particular, illustrates how macroscopic mechanical effects, such as damping and friction, can obscure the underlying thermodynamic behavior of the gas. This underscores the importance of carefully separating idealized theoretical models from real experimental implementations.

11.4 Overall Assessment

Overall, the experiment demonstrates that the determination of thermodynamic quantities such as the heat capacity ratio depends strongly on the chosen measurement method. Among the investigated approaches, the Lissajous phase method provides the most reliable and physically meaningful result within the scope of this experiment. The transit-time and resonance tube methods, while valuable for illustrating alternative principles, are more susceptible to systematic errors and therefore require further refinement to achieve accurate results.

The combined analysis of all tasks highlights the importance of method comparison in experimental physics and provides valuable insight into the strengths and limitations of different experimental techniques.

12 Conclusion

In this experiment, the heat capacity ratio $\gamma = c_p/c_v$ of air was investigated using several independent experimental methods. The speed of sound in air was determined using phase measurements (Lissajous figures) and a transit-time technique, and these values were subsequently used to calculate γ . In addition, a direct determination of γ was attempted using the resonance tube method.

The Lissajous phase method yielded a value of γ that is consistent with the theoretical expectation for air within experimental uncertainty, demonstrating the validity of the adiabatic sound propagation model. The transit-time method provided high statistical precision but resulted in a systematic deviation from the expected value, indicating the presence of dominant systematic timing effects. The resonance tube method produced an unphysical result, highlighting its strong sensitivity to calibration accuracy and non-ideal experimental effects.

Overall, the experiment emphasizes the importance of comparing multiple measurement techniques when determining fundamental physical quantities. While statistical precision is desirable, the reliability of an experimental result ultimately depends on the control and understanding of systematic uncertainties. Among the methods employed, the phase-based sound velocity measurement proved to be the most reliable approach for determining the heat capacity ratio of air within the scope of this experiment.

13 References

M. Ziese, W18e Heat Capacity Ratio , University of Leipzig, January 2026.